Benhard Leroy Situmorang

Analyze the Trend of Inflation Rate in Indonesia from 2020 to 2025 Using ARIMA on Time Series Analysis

Universitas Matana

9% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text

Match Groups

41 Not Cited or Quoted 9%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

7% Internet sources
5% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **41** Not Cited or Quoted 9%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

7% Internet sources
5% Publications
0% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	fastercapital.com	<1%
2	Publication	Ira Safitri, A Rafi Paringgom Iwari. "Forecasting the Inflation Rate in Lampung Pr...	<1%
3	Internet	openknowledge.worldbank.org	<1%
4	Internet	ejournal.iainpalopo.ac.id	<1%
5	Internet	medium.datadriveninvestor.com	<1%
6	Internet	hal.science	<1%
7	Internet	pushstg.indiatimes.com	<1%
8	Internet	www.fastercapital.com	<1%
9	Internet	ojs3.unpatti.ac.id	<1%
10	Internet	bulletin.bmeb-bi.org	<1%

11	Internet	
www.upgrad.com		<1%
12	Publication	
Bishal Das, Priom Seal, Mrinal Saikia, Achintya Pran Hazarika. "Spatial econometri...		<1%
13	Internet	
journal.rescollacomm.com		<1%
14	Internet	
hal.umontpellier.fr		<1%
15	Internet	
link.springer.com		<1%
16	Internet	
medium.com		<1%
17	Internet	
scholar.ub.ac.id		<1%
18	Internet	
www.diva-portal.org		<1%
19	Publication	
Azlan Abdul Aziz, Marina Yusoff, Wan Fairos Wan Yaacob, Zuriani Mustaffa. "Repe...		<1%
20	Internet	
ebin.pub		<1%
21	Publication	
Miguel Esteban, Hiroshi Takagi, Takahito Mikami, Aretha Aprilia, Daisuke Fujii, Sh...		<1%
22	Publication	
Ton Duc Thang University		<1%
23	Internet	
malque.pub		<1%
24	Internet	
r-statistics.co		<1%

25 Publication

Kamini Shahare, Dipanshu Naware, Arghya Mitra, Ritesh Keshri. "Smart load fore... <1%

26 Publication

Nurakmal Ahmad Mustaffa, Siti Mariam Zahari, Nor Alia Farhana, Noryanti Nasir,... <1%

27 Internet

statisticswithr.netlify.app <1%

28 Publication

Abrar Islam, Amanjyot Singh Sainbhi, Kevin Y. Stein, Nuray Vakitbilir et al. "Chara... <1%

29 Publication

Gnanadarsha Sanjaya Dissanayake, Hassan Doosti. "Long Memory Time Series An... <1%

Analyze the Trend of Inflation Rate in Indonesia from 2020 to 2025 Using ARIMA on Time Series Analysis

Benhard Leroy Situmorang[1*], Amanatullah Pandu Zenklinov[2], Reza Septian Pradana[3]

[1,2]*Department of Statistics, Matana University, Tangerang, Banten, Indonesia*
[3]*Badan Pusat Statistik, Tangerang, Banten, Indonesia*
**Corresponding author email: benhard.situmorang@student.matanauniversity.ac.id*

Abstract: Inflation is a key indicator for a healthy, stable, and growing economy that influences the country's purchasing power and monetary policy. In developing or developed economies, a stable inflation rate with a similar wage increase will contribute to a net positive for the people and their purchasing power, so that people can buy the necessary things without being unaffordable. However, internal or external shocks can affect the consumer prices balance, leading to volatile changes in the inflation rate, especially in Indonesia around 2020-2025, with situations like Covid-19 and modern wars. This research aims to analyze the movement of the inflation rate in Indonesia from Bank Indonesia from 2020 to 2025 using the Time Series Analysis method to identify the underlying patterns, seasonal effects, trends, and long-term effects of the Indonesian economy going forward, and future forecast inflation movements using Autoregressive Integrated Moving Average (ARIMA). The ARIMA model will help provide a significant framework for this research, so that we understand the trendline movement and any fluctuations of the inflation rate in Indonesia.

Keywords: ARIMA, Economy, Forecast, Indonesia, Inflation, Time Series

1. Introduction

Inflation is a general increase in the prices of goods and services over a period of time. In its consensus, inflation basically reflects the imbalance between the supply and demand in the national economy, which influences the inflation rate at a national or regional level on the success of economic growth. A high inflation rate can cause higher prices or fluctuations in goods and services, which will cause economic instability. Inflation can be caused by various factors, such as increased demand for products and services (demand-pull), increased production costs (cost-push), internal factors (fiscal and monetary policies), and external factors. Because it affects the prices of goods and services that people are paid for relative to their wages, it's crucial to keep it at a low and stable rate so that prices don't rapidly increase relative to monthly wages (Saputra, 2025).

Indonesia is a major emerging country with fast economic growth and opportunities. Significant factors that affect the Indonesian economy and growth are the domestic interest rate, the amount of money available, and the currency value. For the continuation of economic growth, a low and stable inflation is essential for a fast economic growth rate and an increase in human development, where prices should be on par with the increase in average wages in Indonesia. The maintenance of the amount of money circulation, the supply and demand of

Commented [AZ1]: 2025

Commented [AZ2]: Coba ubah ke kalimat lain/bentuk lain goods and services, and the value of the currency are some of the government and central bank goals in ensuring economic stability. (Novianda, 2025).

The inflation rate in Indonesia has remained low and stable, recorded at 1,95% in April 2025 and 2.31% in August 2025. The success of inflation control from the central bank (Bank Indonesia) has kept the Indonesian economy fast and resilient amidst the high global volatility and uncertainty. (Achmad Reyhan, Bambang Indra Ismaya et al., 2025). However, price stability has always been a major concern for policymakers, practitioners, and researchers because of the important effects on the overall growth of the economy. (Cut Endang Kurniasih, 2024). This is true in Indonesia, which has experienced persistent problems keeping the inflation rate stable in the past, due to the amount of money in circulation and exchange rate fluctuations, which can cause the inflation rate to fluctuate, making it unstable for investment and economic growth. This event has been experienced before in the Asian Financial Crisis 1997-99, where inflation soared to 77,63%. This external shock led to a huge recession in Indonesia, with levels down to -13% in GDP growth in 1998. At the time, hyperinflation did not occur because of the economic stabilization policies in Indonesia. Instead, the lack of a target rate is known as the Inflation Targeting Framework (ITF) to prevent hyperinflation. After this economic crisis, inflation control policies and targets received authorization from Bank Indonesia (Indonesia's central bank) for the maintenance of a low and stable inflation rate at a national and regional level. (Kusumatrisna, 2022).

The analysis of inflation trends is important for Bank Indonesia for the calibration of its interest rates and fiscal and monetary policies, and for the government to manage fiscal interventions and address price volatility. Due to this reason, many researchers and economists use the time series model for the forecast economic variables like inflation. Among various time series models, the ARIMA (Autoregressive Integrated Moving Average) is the most effective model. It has the ability to model historical data patterns and produce a reliable short-term forecast rate. This model combines autoregressive (AR), integrated (I), and moving average (MA) components for the analysis of stationary data and the forecast of errors for short-term predictions (Saputra, 2025).

Commented [AZ3]: Coba ubah ke kalimat lain/bentuk lain

Commented [AZ4]: Ubah ke bentuk baku

The advantage of using ARIMA over other time series models is that it offers high-accuracy results for short-term forecasts, robust handle from linear type time series data, and requires only a single variable to function, which works well on short-term time series data like the inflation rate. It can capture linear trend and seasonality trend that can create patterns and fluctuations of the movement over different timelines. Moreover, the ARIMA method also uses the differencing (d) method, which is used to transform variability data into a stable, stationary data set that can be used to forecast data, which is rarely stationary for inflation data. Based on that consideration, this model can be used to analyze inflation data in Indonesia over the next 5 years, so that it can better look at the inflation situation in the later months (Rassiyanti, 2026).

2. Literature Review

2.1 Understanding Inflation and Its Effects

Inflation is a key macroeconomic indicator for the measurement of economic stability and growth. High inflation has a frequent association with lower growth, currency devaluation, and financial crises. Further price increases are linked to weaker investor confidence, a reduction of incentives to save, and the reasons of financial and public welfare. Most of the negative effects of high inflation fall disproportionately on poor households, because these households are reliant on their wage income for basic needs, have less access to the internet and electricity, and have fewer significant savings or assets. Because of this, a low and stable inflation has been associated with better growth and development outcomes (Ohnsorge, 2019).

2.2 The Inflation-Economic Growth Links in Indonesia

Most of the researchers and economists suggest that there is a negative long-term relationship between inflation and economic growth, particularly in Indonesia. This is because inflation is significantly affected by money supply, supply-demand, interest rate, and exchange rate, which are also indicators for economic growth. Higher inflation can hamper business growth and purchasing power parities, which consequently decreases the total output growth in Indonesia. Because of this, controlling the inflation rate at a low level is crucial for economic growth. (Egi, 2025).

2.3 Time Series Analysis of Inflation Rate

Time series analysis is a statistical method that forecasts future trends from historical data. It's widely used for various domains, such as social, finance, environmental, and economic research, to analyze the patterns, trends, and relationships that affect the movement at any given time. In the context of inflation, time series models help examine the rate of inflation in a specific timeline, from daily, weekly, monthly, or even yearly inflation rates. (Putra, 2026).

The study from AJEBA (2025) uses this method from the inflation data in Indonesia from the World Bank, which spans from 1985 to 2003, where the inflation is proxied by the Consumer Price Index (CPI). It applies the Autoregressive Distributed Lag (ARDL) and the Toda-Yamamoto Test to examine and understand the relationship between inflation and economic growth. These methods together serve as proof that time series analysis is essential for understanding the inflation trends in Indonesia in each timeline (Egi, 2025).

Commented [AZ5]: Coba ubah ke kalimat lain/bentuk lain

Commented [AZ6]: Tambahkan kalimat yang menjelaskan arima lebih baik dari metode lain. Sertakan sumber

One of the main time series, the ARIMA model, is the most widely used in forecasting financial data, because in some cases, this model produces better estimates compared to other forecast methods, such as moving average or autoregression, which is the combination of both in the ARIMA model. (Wulandari et al., 2025).

2.4 ARIMA Method on Inflation Rate

The ARIMA Model, also known as the Box-Jenkins Model, is one of the most widely used statistical methods in time series data. This method combines autoregressive (AR), integrated (I), and moving average (MA) to model patterns from historical data to generate a short-term forecast. The general mathematical form of ARIMA is

$$Y_t = c + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_j$$

The ARIMA method is effective to use on both stationary and non-stationary data, because it can capture trends, seasonality, and autocorrelation structures within observations over time. ARIMA is generally represented as ARIMA (p, d, q), where p represents the autoregressive model, d represents the degree of differencing, and q represents the moving average order (Safitri & Iwari, 2025). This model has received wide adoption in various studies for the forecast of inflation in many countries due to its interpretability and its capacity to handle various forms of data (Wulandari et al., 2025).

3. Materials and Methods

3.1. Materials

3.1.1. Research Method

This type of research method used for this analysis is the quantitative research method. This method is used on numerical data, such as the inflation rate, to help identify distinct seasonal fluctuations, analyze patterns, and forecast data in the future.

The analysis uses the monthly inflation percentage rate in Indonesia from January 2020 to December 2025. It was sourced in Bank Indonesia (BI), the central bank of Indonesia, as it monitors and publishes inflation data monthly. One can locate the Bank Indonesia data on their website, select the range of the period, and download the Excel File.

Table 1. Monthly Inflation in Indonesia from 2020 to 2025

Year	Approximate Inflation Rate
2020	1,32% - 2,98%
2021	1,33% - 1,87%

2022	2,06% - 5,95%
2023	2,28% - 5,28%
2024	1,55% - 3,05%
2025	-0,09 % - 2,92%

Those variation of inflation movements each year were due to external and internal factors that affect the movement. The range of the timeline is used to help create a more effective and accurate short-term forecast, with a shorter timeline than a longer one that requires deeper methods and high computational demand, which can be time-consuming.

3.2. Methods

The method used for this analysis is the ARIMA (Box-Jenkins) model to help identify complex patterns or fluctuations in past inflation rate movements for future forecasting. It involves three stages of the Box-Jenkins approach, which are:

1. Model Identification
2. Modal Estimation
3. Data Forecasting

Those stages became the optimal way of forecasting data. To do the ARIMA method on the data (Inflation Rate in Indonesia from January 2020 to December 2025), these are the steps to create a result with the ARIMA model, these include:

1. Preprocessing Data
2. Stationary Data (ADF-Test)
3. Differencing Data
4. Model ARIMA Identification
5. Parameter Estimation
6. Data Forecasting
7. Evaluation Model

3.2.1. Preprocessing Data

This is the essential part of the data exploration and visualization to transform raw, complex data into patterns that are recognizable and easy to understand, so that we can see what type of data we want to use for our own or for research. To visualize the inflation rate in Indonesia, plot the data to identify monthly trends and fluctuations with the line chart, which is preferable for time series data. Use the line plot with:

X = Monthly Periode
Y = Inflation Rate (%)

to create the initial plot of the inflation data in Indonesia from January 2020 to December 2025. This will help see where the inflation rate is in each month, visualize the movement and patterns, and see any sort of increase, decrease, or fluctuation in a specific timeline.

3.2.2. Stationary Test (ADF Test)

The ARIMA method requires the data to be stationary. While the ARIMA method can deal with non-stationary data, it can't effectively create a stable pattern that is needed for future forecasting, because it has a high level of variance or fluctuation that are difficult to create an accurate model for ARIMA. To check if the data is stationary, use the Augmented Dickey-Fuller Test (ADF-Test) and see if the p-value < 0.05. The hypothesis of the test is

- Null Hypothesis (H_0): $The\ data\ is\ non-stationary$
- Alternative Hypothesis (H_1): $The\ data\ is\ stationary$

If the result were the null hypothesis (where p-value < 0,05), then the data is stationary; if not (where p-value > 0,05), use the differencing method until the data is stationary.

Commented [AZ7]: Kelebihan spasi

3.2.3. Differencing Data

If the data is non-stationary, apply the differencing method to transform the non-stationary data into a stationary series. This works by the removal of any trends and fluctuations that vary the inflation movement and move the rate into an expected range of the rate. The differencing method formula is $y'_t = y_t - y_{t-1}$. This method will make the mean of the time series constant over time, which is essential for accurate data forecasting. If the data isn't stationary, apply the second differencing if needed for stationary data in the next step.

3.2.4. Model ARIMA Identification

If the data is stationary (whether it was by differencing or not), we can move to the ARIMA model identification. This step determines the orders of autoregressive (p), differencing (d), and moving average (q) of the data. Since the d parameter was decided before, this step determines the p and q parameters with the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, where:

- AR (p): Lags in PACF.
- MA (q): Lags in ACF.

When the AR (p) and MA (q) parameters have been decided, it will create an ARIMA (p, d, q) model that can be used for the next step.

3.2.5. Parameter Estimation

Once the ARIMA model was created, the next step was to build the ARIMA model from the plot ACF and PACF and estimate the parameters from the model with Maximum Likelihood Estimation (MLE) to find the most plausible parameter values (significant coefficients) that maximize the likelihood of observations at any given time series data.

3.2.6. Data Forecasting

Once the parameter has been estimated, the ARIMA model can be used to forecast data in the future. Use the fitted ARIMA model from past steps to compute the prediction interval in the future with a specific timeline, like a year (12 months).

3.2.7 Evaluation Model

If, for some reason, the model contains residuals that don't fit the forecast result you would've expected, evaluate the model with the Root Mean Square Error (RMSE), rearrange it, and forecast the data without the residuals in the model.

4. Results

4.1 Line Plot of the Monthly Inflation Rate in Indonesia

To understand more about the complex movement of the inflation rate around that period, the line plot movement of the monthly inflation rate timeline in Indonesia from January 2020 to December 2025 was as follows:

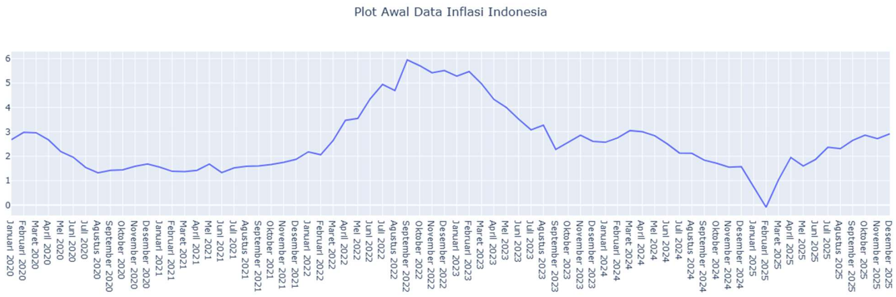


Figure 1. Historical Timeline on the Monthly Inflation of Indonesia (2020 – 2025)

Based on the data from Figure 1 above, we see a series of movements and fluctuations in inflation in Indonesia, with the x-axis as monthly period and the y-axis as inflation rate. The monthly inflation rate begins in January 2020 with a rate of 2,68%. Early in that period, the rate of the index experienced a sharp decline. This is consistent with the decrease in purchasing power of consumers due to worker layoffs, business and shop closures, and a decrease in demand for goods and services both internally and externally (Rahmayani et al., 2021). Then it starts to stabilize in 2021 after a gradual decline in the effects of Covid-19, before the fluctuation increase in 2022 due to the shift in macroeconomic parameters, external conflicts, and supply adjustment. After that, the inflation gradually declined around September 2022 until around the 2 to 3% range, where the rate is stable in the continued months of 2023 to 2025, with a couple of fluctuations in the movement.

4.2 Stationary Inspection and Differencing Data

To see if the data is stationary, an Augmented Dickey-Fuller Test (ADF Test) was conducted on the raw data level. (y_t). The result of the ADF Test without differencing is as follows:

Table 3. ADF Test Result

Statistics	Result
ADF Statistics	-2.713143
P-Value	0.071791
Critical Value (5%)	-2.906
Result	Non-Stationary (Fail to Reject H_0)

Because the p-value (0.071791) was above 0.05 from Table 3, the null hypothesis from this data was not rejected, which means this data was confirmed to be non-stationary. The ARIMA model needs stationary data in order to work, so the first-order differencing (d = 1) was implemented to make the data stationary.

After the first order differencing, the ADF Test was reapplied to check if the data is stationary:

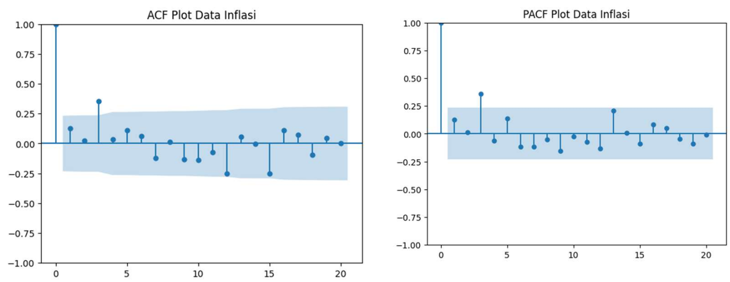
Table 4. ADF Test Result after First Order Differencing

Statistics	Result
ADF Statistics	-2.976025
P-Value	0.037195
Critical Value (5%)	-2.905
Result	Stationary (Reject H_0)

Because the p-value (0.037195) was under 0.05 from Table 4, the null hypothesis from this data was rejected, which means this data was confirmed stationary, which shows the impact of the differencing to remove any trends and fluctuations that carry high variation of the data.

4.3 Model Identification and Estimation

After the stabilization of the data with the differencing method, the ACF and PACF are plotted not only to discover the proper lag structure, but also to identify the ARIMA model used for the estimation of the time series data. Both plots determine the parameter p (PACF plot) and parameter q (ACF plot). The ACF and PACF plots from Figure 2 were created as follows:



(a) (b)

Figure 2. ACF and PACF Plots

Based on both plots from Figure 2, it was revealed that both the ACF and the PACF plot show a significant spike at lag 1, which suggests that the autoregressive parameter (p) is 1 (PACF plot) and the moving average parameter (q) is 1 (ACF Plot). Those are fitted together with the differencing parameter (d) of 1, where the ARIMA (1,1,1) model from the data was identified.

With the ARIMA Model decided, the model estimation from Table 5 was as follows:

Table 5. ARIMA (1,1,1) Model Estimation

Statistics	Coefficient	Standard Error	Z	$P > \|Z\|$
AR Component (ar.L1)	0.774	0.234	3.314	0.001
MA Component (ma.L1)	-0.6476	0.288	-2.247	0.025
Residual Variance (σ^2)	0.1644	0.023	7.122	0.000

Based on the model estimation in Table 5, both AR and MA component levels were below 5% of the p-value (p-value < 5%), which means those components were significant. This confirms that the model was able to capture the linear dependencies and autocorrelations within the inflation time series data and can be suitable for forecasting data.

4.4 Forecasting the Inflation Rate Trend 2026 in Indonesia

The validated ARIMA (1,1,1) model can now be used to forecast the inflation rate movement in Indonesia by 2026, with the forecast result from Table 6 and the visualization from Figure 4 reveals the projected trend and movement below:

Table 6. ARIMA Forecast of the Inflation Rate in Indonesia by 2026

Period (Month)	Inflation Rate
January	2.967%
February	3.003%
March	3.031%
April	3.053%
May	3.0698%
June	3.082%
July	3.093%
August	3.1%
September	3.107%
October	3.111%
November	3.115%
December	3.117%

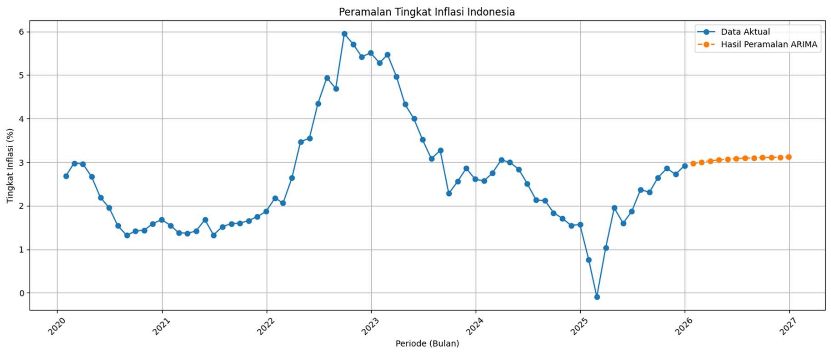


Figure 3. Historical and ARIMA Forecast on the Inflation Rate in Indonesia

The estimated result in Table 6 and the visualization for the forecast in Figure 3 show a slow but gradual increase in the inflation rate. Staring from 2,98% in January 2026, it is expected to reach around 3.12 % in inflation rate by the end of 2026.

4.5 Model Evaluation

To see if the ARIMA model to use for this data is accurate, the model was evaluated with the Root Mean Squared Error (RMSE) for the forecasted inflation rate data.

- RMSE value: 1.44

The RMSE value of 1.44 for macroeconomic forecasting means that the ARIMA (1,1,1) model is considered a good fit for this data, which validates the model's reliability for short-term trend analysis.

5. Discussion

The forecast result of the monthly inflation rate in Indonesia (2020-2025) indicates that the ARIMA model performs very well with the data, at the revelation of the structural behaviors of inflation trends and movements as well as the future short-term trends in Indonesia. It helps to provide several critical insights into the dynamics of Indonesian inflation and the effectiveness of the method itself.

The Augmented Dickey-Fuller Test (ADF-Test) confirms that the data were non-stationary (p-value > 0.05), which is relevant to many characteristics of the macroeconomic time series data. Like many of them, the historical data from 2020 to 2025 reveals a lot of fluctuation and trends that were influenced by internal and external factors, notably Covid-19. This is the reason why the data was non-stationary in the first place, and the application of the first-order differencing was needed to transform the data into a form suitable for the ARIMA model.

After the identification of the ACF and PACF plot show that the ARIMA (1,1,1) model was validated to estimate the models and parameters for the forecast model on the data. The estimation of the ARIMA (1,1,1) model shows that the AR (ar.L1) and MA (ma.L1) components were both significant. The statistical significance implies that the current month's inflation rate for the most part was influenced by past historical rate (autoregressive component) and the previous month's forecast error (moving average component).

The forecast prediction with the ARIMA (1,1,1) model shows that there is a stable but gradual upward trend in the inflation rate in Indonesia. This graph from Figure 3 has grown from 2.98% to 3.12%, which is an increase of 0.14%. But further visualization suggests that the inflation rate will be around 3.1-3.2% range, which still aligns with the expected range from Bank Indonesia. There are several reasons for the forecast result:

1. There is an absence of deflationary pressures in the future compared to before.
2. It indicates a controlled inflationary growth, as the trend is projected to increase 0.15% compared to before, which doesn't signal a huge increase in the inflation rate.
3. It shows the normal rate of the inflation movement, as the recent trend rate in 2025 was lower before that, which suggests that the trend has started to shift in the movement, but not a huge upward trend like hyperinflation.

5. Conclusions

Based on the result of the study and the discussion, it is concluded that the Box-Jenkins ARIMA Model was successfully applied to the monthly inflation rate in Indonesia from 2020 to 2025. It aligns the nature of the ARIMA model on time series, which can still be used on this data, even if it's non-stationary. The application of differencing data, it allows to remove huge variants like fluctuation and seasonal movements to transform it into a stationary data. With that method, it can be used for identification, estimation, and evaluation models needed for forecast data in the short-term, where the ARIMA (1,1,1) model was estimated from those steps. It is estimated that in 2026, the inflation rate is expected to rise slowly but steadily, which offers critical insights for economic decisions for Bank Indonesia or the government.

However, while the ARIMA model was able to forecast the inflation rate based on historical movements and help understand what we expect the inflation rate to move, it doesn't incorporate other variables that can affect the inflation rate. That includes global crude oil prices, exchange rate, money supply, and others. Also, this method only assumes that the statistical measure remains stable, which can change in the distant future with internal and external shocks that the ARIMA method can't account for. Lastly, the historical inflation rate data period from 2020 to 2025 shows the various movements and fluctuations caused by COVID-19 and its aftermath, which showcase that there are still flaws in the accuracy of the forecast model. This is true in the RMSE model of 1.441, which is acceptable, but there were some possible forecast errors in the model. In conclusion, while the ARIMA model works well with the application of the monthly inflation rate in Indonesia and guidance with the forecast model, future research will be needed with deeper and hybrid models for hidden, long-term, and other factors that cannot be applied in ARIMA.

References

Commented [AZ8]: Reference tidak boleh dari web. Gunakan dari paper atau buku

Achmad Reyhan, Bambang Indra Ismaya, C. S. C., Diah Indira, Irham Ilmanel Abdinni, J. A., Kusfisiami Wima Mustika, Nadira Firinda, Nur Azizah, P., Nurman, Rinorsa Duane Agusta, R. B. H. S., & Viyasa Rahyaputra, Wahyunindia Rahman, Y. G. S. (2025). *LAPORAN PEREKONOMIAN INDONESIA 2025*. https://www.bi.go.id/id/publikasi/laporan/Documents/LPI-2025_versi-lengkap.pdf

Cut Endang Kurniasih. (2024). *Unraveling the Dynamic Impact of Money Supply, Interest Rates, and Corruption on Inflation : Evidence from Indonesia*. *22*(June), 1–12. https://doi.org/10.29259/jep.v22i1.23052

Egi, M. (2025). *The Relationship between Inflation and Economic Growth in Indonesia : A Time Series Approach*. *25*(5), 382–393. https://journalajeba.com/index.php/AJEBA/article/view/1809/3989

Kusumatrisna, A. L. (2022). *Bulletin of Monetary Economics and Banking Threshold Effect in the Relationship Between Inflation Rate and Economic Growth in Indonesia*. *25*(1). https://bulletin.bmeb-bi.org/cgi/viewcontent.cgi?article=1886&context=bmeb

Novianda, B. (2025). *Forecasting Inflation in Indonesia Using ARIMAX : The Role of Money Supply and Exchange Rate*. *13*(August), 33–44. https://doi.org/10.33019/equity.v13i2.370

Putra, W. E. (2026). *Implementation of Time Series Forecasting for Inflation Prediction in Indonesia*. *6*(April), 865–874. https://journal.irpi.or.id/index.php/malcom/article/view/2172/1281

Rahmayani, D., Oktavilia, S., & Putri, P. I. (2021). *The Impact of Covid-19 Pandemic on Inflation in Indonesia*. *22*(2), 117–128. https://doi.org/10.23917/jep.v22i2.13861

Rassiyanti, L. (2026). *Modeling and Predicting Indonesia's InflationUsing the ARIMA Model*. *4*(1), 28–34. https://scholar.ummetro.ac.id/index.php/sciencestatistics/article/view/11252/4270

Safitri, I., & Iwari, A. R. P. (2025). *Forecasting the Inflation Rate in Lampung Province Using the ARIMA Method*. *7*(3). https://doi.org/10.32877/bt.v7i3.2335

Saputra. (2025). *Application of Autoregressive Integrated Moving Average (ARIMA) for Forecasting Inflation Rate in Indonesia*. *21*(2), 382–396. https://doi.org/10.20956/j.v21i2.36609

Wulandari, N., Putri, R. A., Nazhirah, A., & Novia, S. A. (2025). *Inflation Forecasting for Riau Province : A Comparison of Parametric ARIMA with Nonparametric Nadaraya-Watson Kernel Regression and B-Spline Methods*. *21*(3), 763–774. https://doi.org/10.20956/j.v21i3.43647